

Comms Overview

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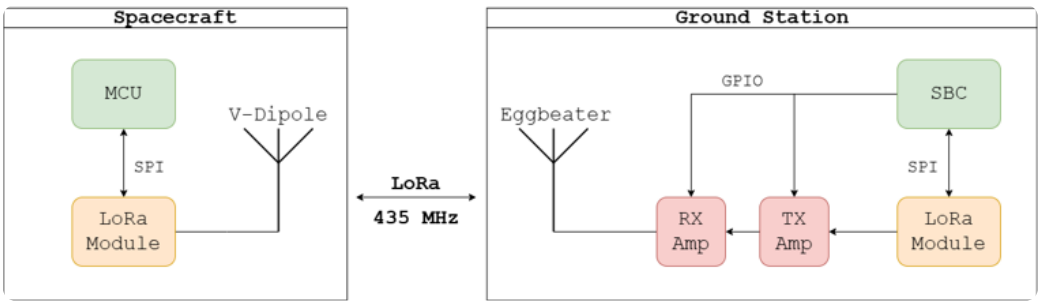
1.0 System Description

Communications Subsystem

The communications subsystem consists of the LoRa communication system (which makes contact with the ground system) and the GPS subsystem.

LoRa Communication Module

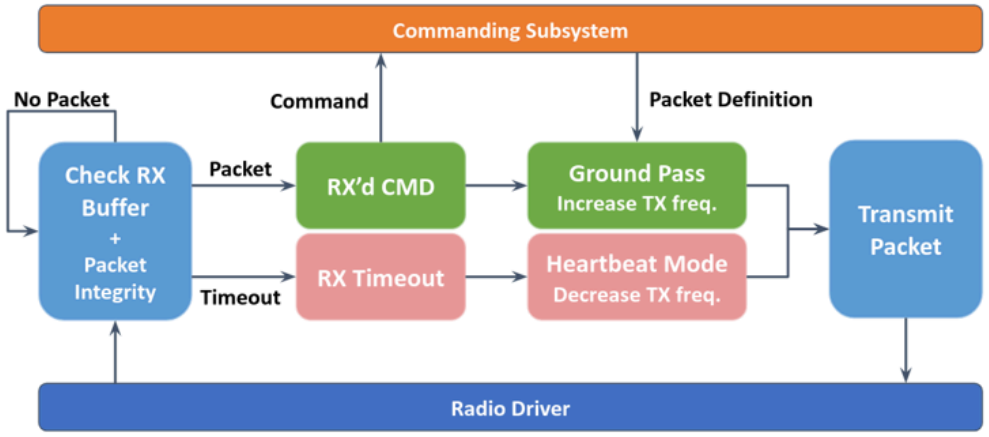
The communications subsystem is responsible for downlinking mission data, establishing and maintaining contact with the ground system, and acting as an interface between the ground system and the rest of the spacecraft. Communications for Argus uses LoRa modulation in the 70 cm UHF band (with a planned center frequency around 435 MHz). The spacecraft's TX power is +30 dBm (1W), and the ground station's TX power is +46 dBm (50W). Through our link budgets, we expect a downlink margin of +12 dB, and an uplink margin of +27 dB for our planned frequency.



Communication Hardware Block Diagram

In orbit, Argus will downlink periodic heartbeats (which we hope to pick up by registering the mission with TinyGS). When a ground pass occurs over our Pittsburgh ground station (GS), the GS shall detect these heartbeats and respond to Argus with uplinked commands. These can request specific mission data or command the spacecraft to perform particular actions.

Downlinking smaller frames of information, such as telemetry frames or ACKs, can be handled in a single LoRa packet. However, for downlinking images and other log files from the OBDH, these files are packetized and downlinked sequentially using a simple protocol, using sequence counts to track packet order (done by the ground system for simplicity). This protocol is completely self-contained within the communications subsystem, acting as a transport protocol for the CDH or other subsystems in the context of downlinking files.



Communications FSW Block Diagram

Global Positioning System (GPS) Module

The GPS subsystem consists of the spacecraft's GPS (Global Positioning System) module, used for general telemetry (gives us position information of Argus in orbit), as well as a ground truth for OD results from the payload.

2.0 Onboarding Information

3.0 Requirements

These requirements are also listed in the Argus requirements document in the Systems Engineering folder (Argus Requirements).

	Requirement	Verification/test method(s)	co-subsystem
1	The system shall have the ability to turn off the space station(s) transmitter(s) immediately, even in the presence of user traffic and/or space station computer system failure. (IARU condition)		
2	The system shall have the ability to ensure the space station is turned off in the event communications are lost with the command station.		
3	Operators shall obtain and provide documentation of proper licenses for the use of radio frequencies per CDS 2.4.1.	Confirm operators have the required licenses.	
4	Per CDS 2.4.2 Argus-2 shall comply with their country's radio license agreements and restrictions.	Confirm documentation of the radio license agreements and restrictions.	
5	The radio transceiver shall operate on the specified bandwidth of the amateur radio frequency bands for CubeSats - UHF, and at the specific frequency that is to be assigned. (433MHz)	Ensure that the transmitted signal is within the allocated UHF band and at the specified frequency. Confirm the signal complies with regulatory requirements for power and bandwidth. Verify that the transceiver operates consistently and stably within this frequency range under nominal and varying conditions.	
6	All deployables such as booms, antennas, and solar panels shall not deploy before a minimum of 30 minutes after Argus-2's deployment switch(es) are	Confirm with the avionics and mechanical team that there are protocols/mechanisms to prevent the release of the antenna prior to 30 min after ejection.	Mechanical/Avionics

	activated during dispenser ejection per CDS 2.4.4.		
7	The CubeSat shall have at least three independent inhibitors to prohibit the inadvertent release of any deployable structures such as antennas or solar panels per CDS 2.3.8.	Confirm with the mechanical team that there are 3 independent methods to prevent the release of antennas.	Mechanical/Avionics
8	Per CDS 2.4.5 the CubeSat shall not generate or transmit a signal earlier than 45 minutes after on-orbit deployment.	Conduct mock deployments before launch and verify with data received from the spacecraft.	
9	The communication system shall support two-way communication, enabling both command uplinks and telemetry, sensor readings, GPS data, and image downlinks.	Conduct a full communication system test by uplinking commands and downlinking telemetry and image data. Ensure that command execution confirmation is received after each command.	
10	The communications subsystem shall achieve a minimum link margin of 3 dB between Argus-2 and the ground station, accounting for free-space path loss, atmospheric losses, antenna gain, and power availability.	Conduct a link budget analysis and achieve a link margin of 3 dB or more.	
11	Argus-2 shall be capable of transmitting and receiving data at a rate of min 1 kb/s (??? update when known) for low data transmission rates and min 10 kb/s (same here) for high data transmission rates.	Attempt to transmit/receive data at a rate of 1 kb/s and 10 kb/s for the duration of one ground pass.	
12	The communications system shall implement a coding scheme (protocol) to ensure	Perform an end-to-end data transmission test by sending known data packets (e.g., test patterns or telemetry data) from the spacecraft's communication system to the	

	data integrity during transmission.	ground station. Introduce artificial noise or signal degradation to simulate realistic space conditions (e.g., low SNR).	
13	The communications team shall develop a data handling protocol that will store downlinked data and be able to display that data on the mission control dashboard.	Confirm data handling and displaying works using mock data.	
14	The communications subsystem shall implement a mission control dashboard to display telemetry data and CubeSat health.	The mission control dashboard correctly displays the received GPS data, altitude, etc.	
15	The team shall develop an operations manual detailing the steps for uplinking commands, downlinking telemetry, and handling contingencies.	Ensure all operational steps (including handling contingencies) are followed as documented. The test should demonstrate that the team can uplink commands and downlink telemetry successfully and respond appropriately to simulated anomalies (e.g., communication blackouts, or command failures).	

4.0 Links to relevant documents

Relevant links to Comms documents, links included.

Software

- Comms Software Reference Documentation (as of `Sep 17, 2025`) - This document list and link to all the files on GitHub that are related to comms software with short description for each of them.

Protocol

Protocol from Argus 1 - may need to be reviewed/updated but good for reference

- Argus-1 Communications Specification
- Comms Message Database

Link Budget

- Argus link budget

TinyGS

- Credentials

BOM

- GS BOM (old - for reference)
- GS BOM

GNSS

- GNSS Notes
- GNSS Trade Study
- Antenna GPS GNSS
- GPS AN0037
- GPS Orion B16 DS
- GPS PATCH CGGBP 18.4.A.02
- PX1122R DS
- S1216F8-GL PB v1
- S1216V8 v0.9

Resources

Some resources for comms: (space comms, study for license, antenna design etc)

- Zac's notes on satellite communication
- Antenna Designs for Cubesats
- Chapter 17 - ARRL
- In-phase and Quadrature Chirp Spread Spectrum for IoT Communications
- No-Nonsense General Class License Study Guide
- Space Communications